Код:

```
#include <linux/module.h>
#include <linux/kernel.h>
#include <linux/fs.h>
#include <linux/rwlock.h>
#include <linux/uaccess.h>
#include <linux/proc_fs.h>
#include <linux/sysfs.h>
#include <linux/string.h>
#include <linux/kobject.h>

static int major = 0; // file devices
static struct proc_dir_entry *test = NULL; // proc fs
static struct kobject *test_kobj = NULL; // sys fs
static rwlock_t lock;
static char test_string[15] = "Hello!\n\0";

ssize_t test_read(struct file *fd, char __user *buff, size_t size, loff_t *off)
{
        size_t rc;
        read_lock(&lock);
        rc = simple_read_from_buffer(buff, size, off, test_string, 15);
        read_unlock(&lock);

        return rc;
}

ssize_t test_write(struct file *fd, const char __user *buff, size_t size,
                loff_t *off)
{
        ssize_t rc;

        if (size > sizeof(test_string) - 1)
                return -EINVAL;

        write_lock(&lock);
        rc = simple_write_to_buffer(test_string, sizeof(test_string), off, buff, size);
        write_unlock(&lock);

        if (rc < 0)
                return rc;

        if (rc < sizeof(test_string))
                test_string[rc] = '\0';
        else
                test_string[sizeof(test_string)-1] = '\0';

        return rc;
}

ssize_t test_proc_read(struct file *fd, char __user *buff, size_t size,
                loff_t *off)
{
        size_t rc = 0;
        rc = simple_read_from_buffer(buff, size, off, test_string, 15);
        return rc;
}
```

```c
ssize_t test_write(struct file *fd, const char __user *buff, size_t size,
                loff_t *off)
{
        ssize_t rc;

        if (size > sizeof(test_string) - 1)
                return -EINVAL;

        write_lock(&lock);
        rc = simple_write_to_buffer(test_string, sizeof(test_string), off, buff, size);
        write_unlock(&lock);

        if (rc < 0)
                return rc;

        if (rc < sizeof(test_string))
                test_string[rc] = '\0';
        else
                test_string[sizeof(test_string)-1] = '\0';

        return rc;
}

ssize_t test_proc_read(struct file *fd, char __user *buff, size_t size,
                loff_t *off)
{
        size_t rc = 0;
        rc = simple_read_from_buffer(buff, size, off, test_string, 15);
        return rc;
}

ssize_t test_proc_write(struct file *fd, const char __user *buff,
                size_t size, loff_t *off)
{
        size_t rc = 0;

        if (size > sizeof(test_string) - 1)
                return -EINVAL;

        rc = simple_write_to_buffer(test_string, 15, off, buff, size);

        if (rc < 0)
                return rc;

        if (rc < sizeof(test_string))
                test_string[rc] = '\0';
        else
                test_string[sizeof(test_string)-1] = '\0';

        return rc;
}

// sys
static ssize_t string_show(struct kobject *kobj, struct kobj_attribute *attr,
                char *buf)
{
        size_t rc = 0;
        memcpy(buf, test_string, 15);
        rc = strlen(test_string);
        return rc;
}
```

```c
static ssize_t string_store(struct kobject *kobj, struct kobj_attribute *attr, char const
*buf, size_t count)
{
        size_t rc = 0;

        if (count > sizeof(test_string) - 1)
                return -EINVAL;

        memcpy(test_string, buf, count);
        rc = strlen(buf);



        return rc;
}

static struct file_operations fops = {
        .owner = THIS_MODULE,
        .read = test_read,
        .write = test_write
};

// kernels 5.* and 6.*
static const struct proc_ops pops = {
        .proc_read = test_proc_read,
        .proc_write = test_proc_write,
};

static struct kobj_attribute string_attribute =  __ATTR(test_string, 0644, string_show, st
ring_store);

static struct attribute *attrs[] = {
        &string_attribute.attr,
        NULL,
};

static struct attribute_group attr_group = {
        .attrs = attrs,
};
```

```c
int init_module(void)
{
        int retval = 0;
        pr_info("Test module is loaded!\n");
        rwlock_init(&lock);
        major = register_chrdev(major, "lab3", &fops);

        if (major < 0)
                return major;
        pr_info("Major number is %d.\n", major);

        test = proc_create("lab3", 0666, NULL, &pops);

        test_kobj = kobject_create_and_add("kobject_test", kernel_kobj);

        if (!test_kobj)
                return -ENOMEM;

        retval = sysfs_create_group(test_kobj, &attr_group);

        if (retval)
        {
                kobject_put(test_kobj);
                return retval;
        }

        return 0;

        /*
        pr_info("Test module is loaded.\n");
        rwlock_init(&lock);
        major = register_chrdev(major, "lab2", &fops);

        if (major < 0)
                return major;
        pr_info("Major number is %d.\n", major);

        return 0;
        */
}

void cleanup_module(void)
{
        unregister_chrdev(major, "lab2");
        proc_remove(test);
        kobject_put(test_kobj);
        pr_info("Test module is unloaded.\n");
}

MODULE_LICENSE("GPL");
```

Сборка:

```
root@kivvi-VirtualBox:~/modules_lab/lab3# make
make -C /lib/modules/6.17.0-35-generic/build M=/root/modules_lab/lab3 modules
make[1]: Entering directory '/usr/src/linux-headers-6.17.0-35-generic'
make[2]: Entering directory '/root/modules_lab/lab3'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
  You are using:           gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
  CC [M]  lab3.o
lab3.c:17:9: warning: no previous prototype for ‘test_read’ [-Wmissing-prototypes]
   17 | ssize_t test_read(struct file *fd, char __user *buff, size_t size, loff_t *off)
      |         ^~~~~~~~~
lab3.c:27:9: warning: no previous prototype for ‘test_write’ [-Wmissing-prototypes]
   27 | ssize_t test_write(struct file *fd, const char __user *buff, size_t size,
      |         ^~~~~~~~~~
lab3.c:50:9: warning: no previous prototype for ‘test_proc_read’ [-Wmissing-prototypes]
   50 | ssize_t test_proc_read(struct file *fd, char __user *buff, size_t size,
      |         ^~~~~~~~~~~~~~
lab3.c:58:9: warning: no previous prototype for ‘test_proc_write’ [-Wmissing-prototypes]
   58 | ssize_t test_proc_write(struct file *fd, const char __user *buff,
      |         ^~~~~~~~~~~~~~~
  MODPOST Module.symvers
WARNING: modpost: missing MODULE_DESCRIPTION() in lab3.o
  CC [M]  lab3.mod.o
  CC [M]  .module-common.o
  LD [M]  lab3.ko
  BTF [M] lab3.ko
Skipping BTF generation for lab3.ko due to unavailability of vmlinux
make[2]: Leaving directory '/root/modules_lab/lab3'
make[1]: Leaving directory '/usr/src/linux-headers-6.17.0-35-generic'
root@kivvi-VirtualBox:~/modules_lab/lab3# ls
lab3.c   lab3.mod    lab3.mod.o  Makefile       Module.symvers
lab3.ko  lab3.mod.c  lab3.o      modules.order
root@kivvi-VirtualBox:~/modules_lab/lab3# 
```

Загрузка:

```
root@kivvi-VirtualBox:~/modules_lab/lab3# insmod lab3.ko
root@kivvi-VirtualBox:~/modules_lab/lab3# dmesg | tail -n 5
[14376.731947] audit: type=1400 audit(1781582407.539:126): apparmor="DENIED" operation="ca
pable" class="cap" profile="/usr/sbin/cupsd" pid=6896 comm="cupsd" capability=12  capname=
"net_admin"
[15630.815576] lab3: loading out-of-tree module taints kernel.
[15630.815582] lab3: module verification failed: signature and/or required key missing - t
ainting kernel
[15630.815950] Test module is loaded!
[15630.815954] Major number is 239.
root@kivvi-VirtualBox:~/modules_lab/lab3# 
```

proc

```
root@kivvi-VirtualBox:/proc# ls -la lab3
-rw-rw-rw- 1 root root 0 Jun 16 00:22 lab3
root@kivvi-VirtualBox:/proc# head lab3
Hello!
root@kivvi-VirtualBox:/proc# printf "test." > lab3
root@kivvi-VirtualBox:/proc# head lab3
test.
root@kivvi-VirtualBox:/proc# 
```

sys:

```
root@kivvi-VirtualBox:/sys/kernel/kobject_test# ls
test_string
root@kivvi-VirtualBox:/sys/kernel/kobject_test# head test_string
test.root@kivvi-VirtualBox:/sys/kernel/kobject_test# printf "qwerty." > test_string
root@kivvi-VirtualBox:/sys/kernel/kobject_test# head test_string
qwerty.root@kivvi-VirtualBox:/sys/kernel/kobject_test# 
```